

KARL HILL

MISSION SOFTWARE
ENGINEERING SYSTEMS | TECHNICAL LEADERSHIP | DELIVERY

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SUMMARY

Staff Aerospace Software Engineer with 20+ years of experience building and delivering production software across national security, aerospace, NASA, and enterprise environments. Combines hands-on engineering with technical leadership across software delivery, engineering practices, mentoring, and cross-team work.

PROFESSIONAL EXPERIENCE

Staff Aerospace Software Engineer

SEPT 2025 — PRESENT

Jacobs National Security · Chantilly, VA

- Lead engineering delivery for a team of about 10 across roughly 20 repositories and multiple deployment environments, sequencing work, coordinating dependencies, and driving integration and release readiness.
- Develop mission software, shared interfaces, messaging integrations, and service orchestration while providing technical guidance and reviewing implementation across the team.
- Advanced a portable messaging layer with a common interface and broker adapters so broker choice can stay in configuration. Ownership is shared.
- Lead Agile planning and execution across internal and partner teams, translating mission priorities into sequenced engineering work and resolving cross-team dependencies and blockers.
- Onboarded and coached approximately six engineers through code review, technical feedback, development guidance, and reinforcement of engineering standards.

Lead Software Engineer

DEC 2017 — SEPT 2025

SSAI / NASA Goddard Space Flight Center · Greenbelt, MD

- Led software engineering on an AWS flood-mapping system for satellite-derived products. GeoHorizons co-author (2026): Software (Equal); Writing – review & editing (Equal). The public map is the shipped artifact.
- Delivered Find Data search, ordering, and near-real-time access for LAADS DAAC, with GitLab CI/CD and Kubernetes web delivery alongside existing archive services.
- Led web engineering on NASA Earth Observatory — editorial workflows, imagery, and the public site. About 1.5 million monthly visitors during that work.
- Automated a content-registry workflow so dataset registration did not depend on a fully manual path.
- Built a Ceph-based file and metadata platform for virtual directory mapping and discovery of large scientific datasets.
- Led Agile software delivery across NASA Earth science teams, coordinating priorities, sprint execution, stakeholder needs, and continuous improvement.

DETAILS

Washington, DC

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LINKS

LinkedIn

linkedin.com/in/khill

GitHub

github.com/karlhillx

Website

karlhill.com

AREAS OF EXPERTISE

- Software Engineering
- Technical Leadership & Direction
- Distributed Systems & Integration
- Engineering Systems & Developer Tooling
- Engineer Development & Coaching
- Agile & Cross-Team Delivery

PROFESSIONAL EXPERIENCE

Senior Software Engineer

JAN 2016 — DEC 2017

InformedDNA · Washington, DC

- Designed and delivered a Laravel case-management platform connecting counseling workflows, documentation, and billing.
- Built CRM enhancements for customer lifecycle workflows and reporting, and tightened maintenance and security processes with operations.

Senior Software Engineer

JUN 2012 — MAR 2015

Ticomix, Inc. · Washington, DC

- Delivered SugarCRM solutions for more than 20 clients, including the Virginia Department of Transportation and Kastle Systems.
- Improved backlog management and delivery practices across client projects.

Earlier Engineering Experience

1997 — 2012

Sabre Corporation · Dante Inc. · Visitar Inc. · Verizon Business

- Held software engineering and principal-level roles across travel, enterprise CRM, telecommunications, and managed security.
- At Sabre, built new PHP applications and added features to existing ones for large-scale travel systems.

AREAS OF EXPERTISE

Software Engineering · Technical Leadership & Direction · Distributed Systems & Integration · Engineering Systems & Developer Tooling · Engineer Development & Coaching · Agile & Cross-Team Delivery

PUBLICATIONS

Peer-reviewed research: A web-based high-resolution global water and flood mapping platform. Co-author, A web-based high-resolution global water and flood mapping platform, GeoHorizons (AGU / Geological Society of London), 2026. Software engineering for a NASA-supported system delivering near-real-time, high-resolution satellite-derived water and flood products worldwide. Author contribution (CRedit): Software (Equal); Writing – review & editing (Equal). doi:10.1144/gh2025-7

EDUCATION

Computer Science coursework, University of Maryland, Baltimore County

Associate of Arts, General Studies — Computer Science Curriculum, Howard Community College

Project Management Certificate, Rutgers University

CERTIFICATIONS

SAFe Agilist, Scaled Agile

Professional Scrum Master II, Scrum.org

Professional Scrum Product Owner I, Scrum.org

OPEN SOURCE

The Dry Standard — Production non-alcoholic drink catalog & comparison engine; open-source codebase.

bb-run — Run Bitbucket Pipelines locally.

testrisk — Rank the highest-value Python test gaps from coverage, AST, and git.

TECHNICAL EXPERTISE

Languages: Python, TypeScript, Java, Bash

Cloud & Containers: AWS, Docker, Kubernetes, Helm

CI/CD & Developer Tooling: Git, Bitbucket Pipelines, GitLab CI, GitHub Actions, release automation, security scanning

Python Tooling: uv, Ruff, ty, pytest, pre-commit — dependency management, linting, type checking, testing, and automated quality gates

Integration: REST/OpenAPI, RabbitMQ, ActiveMQ, distributed systems

Data: PostgreSQL, MySQL, SQL Server, Elasticsearch/OpenSearch, Redis